

PREDICTING UNIVERSITY ENROLLMENT DYNAMICS IN CHILE USING MACHINE LEARNING: A REPLICABLE FRAMEWORK WITH MULTI-SOURCE INTEGRATION AND TEMPORAL VALIDATION

PREDICCIÓN DE LA DINÁMICA DE MATRÍCULA UNIVERSITARIA EN CHILE MEDIANTE MACHINE LEARNING: UN MARCO REPLICABLE CON INTEGRACIÓN MULTIFUENTE Y VALIDACIÓN TEMPORAL

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Abstract

This innovation paper describes a replicable machine learning framework for forecasting university enrollment at fine granularity (institution × program × region × schedule × gender × student status × year) in the Chilean higher education system. Most existing studies in Latin America rely on nationally aggregated data, classical econometric models without strict temporal validation, or report inflated metrics due to random splits that leak future information into training. This work addresses these gaps by integrating ten official data sources (SIES, CNED, INE, Central Bank, mifuturo.cl) into a consolidated dataset of 63,165 observations covering 2018–2025. An expanding window cross-validation strategy with fold-aware target encoding is applied to ensure temporally honest evaluation. Three gradient boosting models (LightGBM, XGBoost, CatBoost) are optimized using Optuna, alongside four informed baselines. The winning model (CatBoost) achieves $R^2 = 0.944$ and a weighted absolute percentage error (WAPE) of 15.0% on the 2025 holdout at fine granularity, improving 49.5% over the naïve baseline. National-level aggregated accuracy reaches 99.4%. Additionally, three methodological contributions are provided: (i) empirical quantification of bias introduced by incorrectly applied target encoding (2.0% RMSE inflation); (ii) an automated method for detecting three types of pandemic shock (level, dispersion, relapse); and (iii) a 2026 national enrollment forecast of 612,578 students (+2.60% over 2025).

Keywords: university enrollment, machine learning, gradient boosting, temporal validation, Chilean higher education

Resumen

Este artículo de innovación describe un marco replicable de machine learning para la predicción de matrícula universitaria a granularidad fina (institución × carrera × región × jornada × género × condición × año) en el sistema de educación superior chileno. La mayoría de los estudios existentes en América Latina trabaja con datos agregados a nivel nacional, utiliza modelos econométricos clásicos sin validación temporal estricta, o reporta métricas infladas por particiones aleatorias que filtran información futura al entrenamiento. Este trabajo aborda dichas brechas integrando diez fuentes oficiales (SIES, CNED, INE, Banco Central, mifuturo.cl) en un dataset consolidado de 63.165 observaciones para el período 2018–2025. Se aplica una estrategia de validación temporal con expanding window cross-validation y target

encoding fold-aware para garantizar evaluación temporalmente honesta. Se optimizan tres modelos de gradient boosting (LightGBM, XGBoost, CatBoost) con Optuna, junto con cuatro baselines informados. El modelo ganador (CatBoost) alcanza $R^2 = 0,944$ y un error porcentual ponderado (WAPE) de 15,0% sobre el holdout 2025 a granularidad fina, mejorando un 49,5% sobre el baseline naïve. La precisión agregada nacional alcanza 99,4%. Se aportan además tres contribuciones metodológicas: (i) cuantificación del sesgo por target encoding mal aplicado (2,0% de inflación del RMSE); (ii) un método automatizado de detección de tres tipos de shock pandémico; y (iii) un pronóstico de matrícula nacional 2026 de 612.578 estudiantes (+2,60%).

Palabras clave: matrícula universitaria, machine learning, gradient boosting, validación temporal, educación superior chilena

1. Introduction

Enrollment forecasting is a critical input for institutional planning and public policy in higher education systems. At the macro level, policymakers need national projections to anticipate subsidy demand and assess the sustainability of student loan programs. At the meso level, universities require program-level forecasts for budgeting, infrastructure sizing, and retention planning. At the micro level, admissions and academic teams need granular information for cohort management and marketing decisions.

However, predicting university enrollment is structurally difficult for at least three reasons. First, the Chilean university system generates signals from heterogeneous sources that are rarely modeled jointly: enrollment statistics from SIES (MINEDUC), institutional indicators from CNED, regional socioeconomic variables from INE and the Central Bank, employability metrics from mifuturo.cl, and accreditation data from CNA. Second, enrollment exhibits structural heterogeneity across multiple simultaneous dimensions: institution, program, region, schedule, gender, and student status. Nationally aggregated models lose this heterogeneity, while fine-grained models face elevated idiosyncratic noise. Third, the 2020–2022 period introduced an unprecedented structural anomaly—the COVID-19 pandemic—that affected enrollment differentially each year, requiring explicit treatment in any predictive model aspiring to future generalization.

The applied literature on enrollment forecasting in Latin America presents several recurring limitations. Most studies operate on nationally or institutionally aggregated data, losing program-level heterogeneity. When machine learning is employed, random splits are frequently used, leaking future information into training and artificially inflating reported metrics. Pandemic effects are typically either ignored or treated with binary dummies without quantitative justification.

This work addresses these gaps through four contributions: (C1) a replicable framework integrating ten official Chilean sources at fine granularity, producing a consolidated dataset of 63,165 observations; (C2) empirical evidence that incorrectly applied target encoding inflates apparent performance by 2.0% RMSE; (C3) an automated pandemic shock detection method identifying three distinct shock types; and (C4) a temporally validated predictive model achieving $R^2 = 0.944$ on the 2025 holdout with a national enrollment forecast for 2026.

2. Materials and methods

2.1 Data sources and integration

Ten official sources from the Chilean higher education ecosystem were integrated: SIES enrollment microdata (MINEDUC), CNED BaseINDICES (tuition fees, admission scores, program duration), CNED faculty composition and infrastructure data, Central Bank regional GDP, INE regional unemployment rates, mifuturo.cl retention and employability indicators, and three static dictionaries mapping institutional codes, program names, and campus-to-region assignments. The integration pipeline comprised five sequential phases: (i) compatibility auditing across sources to identify encodings (UTF-8 with BOM, Chilean locale

separators) and native granularities; (ii) key normalization using canonical region, institution, and program dictionaries; (iii) sequential merging on common keys (year + region, year + institution + program); (iv) hierarchical imputation across five cascading layers; and (v) feature engineering producing 30 derived variables.

The final dataset contains 63,165 unique observations at institution × program × region × schedule × gender × student status × year granularity, covering 2018–2025. The universe is restricted to universities (56 institutions, approximately 600,000 students annually). Rows with prior-year enrollment below 10 students were excluded to reduce noise from marginal programs.

2.2 Target variable

The target is the enrollment growth rate: $(\text{enrollments}_t - \text{enrollments}_{t-1}) / \text{enrollments}_{t-1}$, transformed using the inverse hyperbolic sine function (arcsinh) to compress tails while preserving sign and symmetry. Predictions are reverted via the hyperbolic sine function and reconstructed into absolute enrollment numbers. Winsorization at the 1st and 99th percentiles was applied prior to transformation.

2.3 Hierarchical imputation

Missing values were imputed through five cascading layers, each operating only on cells left empty by the previous layer: temporal interpolation for faculty and infrastructure variables (4.3% of cells); hierarchical grouping by institution-program, institution-area-year, and area-year for tuition and retention variables (18.1%); region-program grouping (11.4%); institution-level grouping (35.3%); and terminal imputation using global medians (30.9%). A binary flag tracks which rows contain any imputed value.

2.4 Temporal validation strategy

The temporal structure of the data prohibits random splits. An expanding window cross-validation was implemented: fold k trains on years $\{2019, \dots, 2018+k\}$ and validates on year $2019+k$, for $k = 1, \dots, 5$. Year 2025 is reserved as a blind holdout. Out-of-fold retrospective predictions were generated for each year by training exclusively on prior data, producing temporally honest predictions for the entire historical period. Critically, target encoding for high-cardinality categorical variables (region, institution, program, knowledge area) was implemented inside the cross-validation pipeline using Bayesian smoothing ($m = 30$), ensuring no information leakage from validation data.

2.5 Models evaluated

Four informed baselines (constant median, historical mean by combination, Ridge with lag only, Ridge with full feature set) and three gradient boosting models (LightGBM, XGBoost, CatBoost) were evaluated. Hyperparameter optimization was performed using Optuna with 50 trials per model, employing Tree-structured Parzen Estimator sampling and median pruning.

3. Findings

3.1 Global model performance

CatBoost emerged as the winning model with an RMSE of 0.3109 in cross-validation on the arcsinh -scaled target, marginally outperforming LightGBM (0.3122) and XGBoost (0.3140). On the 2025 holdout, the final model achieves $R^2 = 0.944$, $\text{MAE} = 11.56$, and $\text{WAPE} = 15.01\%$, translating to 85.0% aggregated confidence. The skill score of +49.45% over the naïve baseline (which simply projects prior-year enrollment) confirms that the model provides substantive predictive value beyond temporal inertia.

Table 1. Global performance of CatBoost model (after tuning with 50 Optuna trials)

Dataset	n rows	RMSE	MAE	R^2	WAPE (%)	Confidence (%)	Skill vs naïve (%)
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OOF retrospective (2019–2025)	54,721	36.64	13.40	0.877	18.36	81.64	+62.13
Holdout 2025	7,752	26.59	11.56	0.944	15.01	84.99	+49.45

3.2 Precision scales with aggregation level

A central finding is that model precision varies systematically with aggregation level. At the finest granularity (institution × program × schedule × gender × condition), WAPE is 15.0% on the 2025 holdout. Aggregating to institution level yields 96.1% confidence (WAPE = 3.88%), and national annual aggregation reaches 99.4% confidence (WAPE = 0.64%). This pattern is theoretically coherent: at higher aggregation, idiosyncratic errors partially cancel. The practical implication is that the model is highly suitable for system-level planning (national, regional, institutional) and moderately suitable for program-specific predictions.

Table 2. Model performance by aggregation level (Holdout 2025)

Level	n groups	WAPE (%)	Confidence (%)	R ²
1. National annual	1	0.64	99.36	n/a
2. + Institution	51	3.88	96.12	0.993
3. + Knowledge area	390	6.27	93.73	0.987
4. + Region	598	6.92	93.08	0.988
5. + Program	2,197	10.48	89.52	0.963
6. + Student status	4,206	12.84	87.16	0.956
7. + Gender	7,363	14.70	85.30	0.946
8. + Schedule (fine)	7,720	15.00	85.00	0.944

3.3 Year-by-year performance and pandemic impact

The retrospective out-of-fold analysis reveals three noteworthy patterns. First, R² increases steadily in post-pandemic years, from 0.841 in 2023 to 0.944 in 2025, reflecting improved learning with more training data. Second, R² drops to 0.674 in 2020 due to the structural discontinuity of the pandemic, but recovers gradually. Importantly, even during 2020, the aggregated national error remains manageable at +6.80%. Third, the year 2019 achieves R² = 0.921 despite having the smallest training set (only 2018 as prior data), confirming that the pandemic anomaly—not data scarcity—was the limiting factor during 2020–2023.

The automated shock detection method classified 2020 as a level shock (Z-score on mean = −2.19), 2021 as a dispersion shock (Z-score on standard deviation = +2.04), and 2022 as a relapse shock (negative aggregate variation not captured by Z-scores). Three binary dummy variables were added to the feature set accordingly.

3.4 Target encoding bias quantification

A controlled experiment comparing fold-aware target encoding against a naïvely precalculated version (computed over the entire dataset before splitting) revealed that the leaky approach reports RMSE 2.0% better than the model would actually achieve in production (0.3047 vs. 0.3109). While modest in absolute terms, this bias can invert the apparent ranking of competing models in applied studies where differences are typically in the 1–3% range.

3.5 SHAP-based interpretability

SHAP analysis reveals that the prior-year enrollment lag (matriculas_lag1) and the new-vs-continuing student indicator (CONDICION_NUEVO) are the two most influential features, with nearly equal importance. This reflects a structural reality: enrollment dynamics for new students (driven by admissions

and recruitment) are fundamentally different from those of continuing students (driven by retention). Program name, regional market share, and schedule type round out the top features.

3.6 National forecast for 2026

The model projects national enrollment of 612,578 students for 2026, equivalent to +2.60% growth over 2025. This forecast is consistent with simple baselines based exclusively on post-pandemic years (differences below 0.5 percentage points), supporting its reasonableness, while diverging substantially from a linear regression over the full 2018–2025 period (which predicts –3.64% due to incorporating the 2020 shock).

4. Implications

Theoretical implications. This work contributes to the literature on machine learning applied to higher education in three ways. First, it provides the first published benchmark for fine-grained enrollment prediction in Chile (and, to the authors' knowledge, in Latin America), establishing performance baselines at eight levels of aggregation that future studies can use for comparison. Second, it provides empirical quantification of target encoding bias—a methodological concern frequently mentioned but rarely measured. The finding that precalculated target encoding inflates RMSE by 2.0% constitutes a concrete call for methodological rigor in applied machine learning studies that employ this technique on high-cardinality categorical variables. Third, the automated pandemic shock detection method offers a replicable alternative to the ad hoc dummy treatments prevalent in the literature, enabling differentiated classification of level, dispersion, and relapse shocks based on quantitative criteria rather than researcher judgment.

Practical implications. For policymakers, the model provides a national-level projection tool with 99.4% confidence, directly applicable to subsidy planning, student loan sustainability assessment, and system-wide capacity dimensioning. For university administrators, the framework enables institutional-level enrollment forecasting with approximately 96% confidence, supporting budgeting, faculty hiring, and infrastructure investment decisions. At the program level, approximately 89% confidence allows anticipating which programs may face enrollment declines, informing decisions about cohort openings, program closures, and targeted marketing campaigns.

The modular pipeline architecture—with independent ETL notebooks per data source and a final merge pipeline—is designed for maintainability: as new annual data becomes available, individual pipelines can be updated independently without restructuring the entire workflow. The exclusive reliance on publicly available official data sources ensures that any Chilean institution can replicate and adapt the framework to its specific needs.

Finally, the distinction between rate-level and absolute-level performance metrics has practical implications for how enrollment forecasting studies communicate their results. Studies reporting metrics only on growth rates may substantially underestimate the practical utility of their models for planning audiences, who ultimately need accuracy in absolute enrollment numbers.

5. Conclusions

This work proposed a replicable machine learning framework for predicting university enrollment in Chile at fine granularity (institution × program × region × schedule × gender × student status × year), integrating ten official data sources into a consolidated dataset of 63,165 observations for the 2018–2025 period.

The main findings are fourfold. First, the winning CatBoost model achieves $R^2 = 0.944$ and WAPE = 15.0% on the 2025 blind holdout at fine granularity, scaling to 99.4% confidence at the national aggregate level, improving 49.5% over the naïve baseline. This demonstrates that modern gradient boosting methods, when

properly trained with strict temporal validation, can deliver actionable enrollment predictions for institutional and system-level planning.

Second, the empirical quantification of target encoding bias revealed that precalculating encodings over the entire dataset—a common but methodologically incorrect practice—inflates apparent performance by 2.0% RMSE. This finding carries implications beyond the specific application: any study using target encoding on high-cardinality variables must implement it within the cross-validation pipeline to report honest metrics.

Third, the automated pandemic shock detection method successfully identified three differentiated shock types in the Chilean university system (level in 2020, dispersion in 2021, relapse in 2022), offering a principled alternative to the monolithic treatments prevalent in the applied literature.

Fourth, the 2026 national forecast of 612,578 students (+2.60% over 2025) is consistent with post-pandemic trend baselines, supporting its use as an informed planning input.

Key limitations include the restriction to universities (excluding technical institutions that represent approximately half of higher education enrollment), lower performance in small regions with few observations, and the assumption of macroeconomic stability underlying the 2026 forecast. Future work should extend the framework to the CFT/IP subsector, explore hierarchical Bayesian models for small regions, and develop sensitivity analyses under alternative macroeconomic scenarios.

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